

Thesis Summary

Title: «*Higgs Strahlung via contact interactions through next-to-next-to-leading order QCD*»

The present thesis focuses on the analytical computation of squared amplitudes for the process $q\bar{q}' \rightarrow WH$, followed by the decay of the W boson into a lepton-antilepton pair, within the framework of the Standard Model Effective Field Theory (SMEFT). By treating the Standard Model (SM) as a limiting case of SMEFT, the present study systematically quantifies deviations introduced by higher-dimensional operators, providing a deeper understanding of possible new physics effects. Such studies are essential for testing the validity of the Standard Model and the exploration of potential extensions that could explain phenomena not currently described by known physics.

A significant contribution of this work is the derivation of squared amplitude expressions up to next-to-next-to-leading order (NNLO) in perturbative quantum chromodynamics (QCD). This level of precision is crucial for ensuring theoretical accuracy in collider physics, particularly for experiments conducted at the Large Hadron Collider (LHC). The analytical expressions obtained in this thesis have been rigorously tested for consistency and validated against known results at lower orders and in the SM limits, thereby reinforcing their reliability. Extending calculations to NNLO accuracy is of particular importance in the SMEFT framework, as it allows for a more precise determination of how new physics effects may alter experimental observables.

In addition to purely analytical calculations, numerical verification was performed using a Python tool developed by the author. This tool evaluates squared amplitudes and cross-checks them against established computational frameworks such as OpenLoops and MadGraph. This verification process played a crucial role in confirming the accuracy of the derived expressions and ensuring robustness in the SMEFT framework. Additionally, these numerical techniques provide a practical way for experimental physicists to compare theoretical predictions with data collected from high-energy collisions.

One of the significant findings of this work is the role of CP-odd SMEFT operators, which introduce new interactions that do not interfere with SM amplitudes. This results in unique modifications to the predicted cross sections, rendering these effects experimentally distinguishable. The results demonstrate that NNLO QCD corrections significantly impact these contributions, often in ways that differ from SM corrections. Understanding these effects is vital for identifying potential signals of new physics in LHC data.

The findings of this thesis contribute to the ongoing effort to probe the limits of the Standard Model and search for new physics through high-precision calculations. The developed analytical framework provides a foundation for further phenomenological studies and experimental analyses in SMEFT-based collider physics. By improving theoretical predictions and providing validated computational tools, this research enhances the ability of physicists to interpret collider data and explore potential signs of physics beyond the Standard Model.